Effect of yielding on the thermo-mechanical response of composite sections

Xiaotian Huang

Master of Engineering
Fire Safety Engineering



The University *of* Queensland
2025

Effect of yielding on the thermo-mechanical response of composite sections

by Xiaotian Huang

(Progress Report)

Student Number: **s4813765**
Course Code: **CIVL7513**
Supervisor: **Dr Anwar Orabi**
Submission Date: **09/06/2025**

School of Civil Engineering
The University of Queensland

Acknowledgements

Placeholder, to be completed at the completion of the project.

Abstract

To be completed at end of the project.

Table of Contents

1. Introduction 4
1.1 Background 4
1.2 Research Objectives 4
2. Literature Review 6
2.1 Theoretical Advancements 6
2.1 Fundamentals ..... **Error! Bookmark not defined.**
2.1 Material Non-linearity ..... **Error! Bookmark not defined.**
3. Methodology 9
4. Model Validation 11
4.1 Experimental Data 11
4.2 Numerical Simulation 12
4.2.1. Heat Transfer 12
4.2.1. Thermo-Mechanical Model 15
Bibliography 19
Appendix A 0

Table of Figures

Figure 1: Plot of temperature deflection response from (Usmani et al., 2001) 8

Figure 3: Location of Measurement Instruments (Wang et al., 2017) 11

Figure 4: Temperature-Time Curve of Furnace (Left) and Thermocouples (Right) (Wang et al., 2017) 12

Figure 4: Vertical Displacement to Temperature at Bottom Flange (Wang et al., 2017) 12

Figure 5: Temperature Profile of Heat Transfer Model – Insulation properties as specified in report 14

Figure 6: Nodal Temperature at Thermocouple Locations – Insulation properties as specified in report 14

Figure 5: Temperature Profile of Heat Transfer Model – Increased insulation conductivity 15

Figure 6: Nodal Temperature at Thermocouple Locations – Increased insulation conductivity 15

Figure 7: Deflection Contour of Thermo-Mechanical Model 16

1. Introduction

1.1 Background

A strong and robust structural framework is one of the foremost considerations in designing a building that is resilient against the adverse effects of fire The structure must be capable of withstanding potential fire scenarios and resisting collapse to achieve the primary objectives of life safety, property protection, and the facilitation of first responder intervention.

Standards and regulations concerning structural fire safety—along with general fire safety—have largely evolved in response to catastrophic events involving significant building fires. A notable example is the events of September 11, 2001, which saw the complete collapse of World Trade Centre 7, a steel-framed structure, solely due to fire (Orabi et al., 2020). Such incidents have driven a strong and continuing interest in advancing the fundamental understanding of structural behaviour under fire conditions (Engelhardt & Morovat, 2022).

Current structural codes are primarily based on standardized fire testing of individual elements in isolation. These tests do not account for the holistic behaviour of entire structural systems or the complex interactions among various structural mechanisms during a fire. For example, the forced thermal expansion of horizontal members can induce failure at critical connections, potentially compromising structural integrity. Conversely, highly redundant frames may develop alternative load paths, enabling the structure to withstand severe fire exposure with minimal damage (Rotter et al., 1999).

Research by (Usmani et al., 2001) demonstrated that composite steel-framed structures exhibit significantly greater inherent fire resistance than suggested by tests on isolated steel members in fire furnaces. This reveals potential for economic efficiencies by avoiding overly conservative designs. However, this early work was limited to elastic material behaviour and did not examine the effects of material non-linearity. A more comprehensive understanding of how plastic yielding affects composite steel-concrete elements is necessary to elucidate the mechanisms of failure and the overall response of the structural system.

Given the substantial cost and logistical challenges of conducting full-scale fire tests, numerical simulations using software such as ABAQUS and OpenSees have become essential tools for analysing and predicting the thermo-mechanical response of structures in fire conditions.

1.2 Research Objectives

This project aims to investigate the thermo-mechanical response of composite structural elements subjected to fire, with a particular focus on the effects of plastic deformation due to material yielding. The study will consider a typical composite element consisting of a steel beam and concrete slab, a prevalent structural configuration in modern building design.

The research will explore how temperature-induced thermal gradients and mechanical yielding influence the section capacity and failure mechanisms of the composite element. A key focus will be the comparison between responses predicted using linear (elastic) and nonlinear (plastic) material properties.

Numerical simulations will be carried out using ABAQUS across a range of section sizes to evaluate the structural response under varying thermal loads, boundary conditions, and material behaviours. Through this analysis, the following research questions will be examined:

- What are the dominant failure mechanisms of composite steel-concrete elements under varying thermal and mechanical loading conditions.
- Does the dominant response of the composite element (e.g., thermal expansion versus bowing) change under nonlinear material conditions?
- Does thermal degradation of material properties have significant influence on the section capacity of the composite beam?
- When accounting for thermal degradation and yielding, do the results support or challenge the conclusions of Usmani et al. (2001), which suggest that composite structures possess a high level of inherent fire resistance, potentially leading to overly conservative design practices?

The outcomes of this investigation aim to enhance the understanding of the thermo-mechanical behaviour of composite structural elements under fire conditions. By examining the influence of material nonlinearity, thermal gradients, and thermal degradation, the study seeks to provide deeper insight into the mechanisms governing structural response and failure during fire exposure.

2. Literature Review

2.1 Theoretical Advancements

Historically, the behaviour of composite structures in fire was believed to be primarily governed by the loss of material strength due to thermal degradation and the effects of loading on the weakened structure, which are mechanisms fundamentally similar to those influencing structural response under ambient conditions. However, this view has been challenged by more recent research, which shows that thermally induced forces and the resulting displacements dominate the structure response in fire (Usmani, 2003).

These salient advancements can be attributed to the 1990 Broadgate development fire and the subsequent Cardington frame fire tests in 1995 which provided critical empirical data for further research and insight for structure response to fire (Lamont, 2001). Numerical analyses of the Cardington tests by (Gillie et al., 2002) concluded that the material degradation and gravitational loading were of secondary importance while the structural response was dominated by the effects of thermal expansion.

Despite this advancement in understanding, current structural design practices often neglect the complex interactions between structural elements. In reality, such interactions often lead to a high degree of redundancy, allowing extensive redistribution of loads through alternative load paths. This inherent capacity for redistribution creates sufficient reserve capacity for such structures to endure severe fires without collapse (Rotter et al., 1999).

In structural fire resistance design, it is commonly (though not always accurately) assumed that the fire remains locally contained within the compartment of origin. This allows the simplification of the expected behaviours from such fires and facilitates the prediction and design of the required fire resistance necessary for the structure to maintain its capacity. Outside of this assumption, the behaviour of the structure becomes difficult to predict due to the unknown heat regimes and failure sequence of the building elements.

Much like how composite steel-concrete structures can withstand higher loads, they also have significantly more fire resistance than individual structural members due to their ability to redistribute the loads to parts with a higher stiffness. This phenomenon is supported by displacement numerical models based on the Cardington tests (Rotter et al., 1999). Research by (Bailey et al., 2000) also found that composite flooring systems have greater inherent fire resistance than that indicated by codified design methods due to the development of tensile membrane action in the slab system at high displacements.

Although large and small-scale fire tests, alongside advances in numerical modelling, have significantly improved the theoretical understanding of structural fire behaviour, this body of knowledge remains incomplete and has yet to be fully integrated into a unified framework capable of informing practical and comprehensive design guidance.

2.1 Response Mechanisms in Fire

A structural element subjected to a fire will increase in temperature and subsequently increase in size dependent on its material properties. For a composite beam, this effect is observed in two principal behaviours of thermal expansion and thermal bowing (SFPE, 2016).

Thermal expansion is the elongation of the element which induces compressive forces due to the restrained end conditions. This behaviour is dominate when the structural element experiences uniform heating. The significance of this effect is often underestimated by most

professionals in structural fire design, with studies by (Usmani et al., 2003) concluding that it may be a large contributing factor in the collapse of the World Trade Centre 7 building. Given the structure's slender floor system, restrained thermal expansion can lead to buckling of the floor and degraded axial stiffness resulting in a detrimental effect on its lateral support capacity. However, in conventional steel frame composite structures thermal expansion can also be beneficial by reducing the internal forces experienced by structural elements when heated.

Alternatively, where a thermal gradient is developed within the element due to uneven heat exposure, the portion with a higher temperature experiences greater increases in size and therefore greater thermal elongation. This leads to bending of the element forming the shape of a bow, with the overall element experiencing tensile forces. The steeper the thermal gradient across the section, the greater the bending and tensile forces. This phenomenon is aptly named thermal bowing.

The thermal bowing phenomena is more prominent in materials with a higher coefficient of thermal expansion, i.e. steel, and is therefore an essential consideration and expected behaviour for composite structures affected by fire. Research by (Kowalski et al., 2018) found that for RC (reinforced concrete) beams exposed to non-uniform heating, deflection due to thermal bowing can be 10% to 20% of the total deflections of the loaded RC beams. This percentage can be expected to be even higher for composite steel beams and concrete slabs, where the steel forms a larger proportion of the section and is centralised below the slab.

In consideration of the two thermal behaviours mentioned above, the fundamental relationship for thermo-mechanical response of structures under thermal influence is expressed via the following, where the total strain dictates the deformation of the element (Usmani et al., 2001):

$$\varepsilon_{total} = \varepsilon_{thermal} + \varepsilon_{mechanical}$$

Where the structural element is unrestrained and can develop freely from thermal expansion, there are no mechanical strains resulting in the total strain being equivalent to the thermal strain ($\varepsilon_{total} = \varepsilon_{thermal}$). Conversely when the element is fully restrained, the total deflection and total strain is equal to zero such that a mechanical strain is produced in response to the thermal strains generated ($\varepsilon_{thermal} = -\varepsilon_{mechanical}$). For real scenarios of structures under fire, there will be a combination of mechanical strain from applied loads and mechanical strain in response to restrained thermal expansion, the magnitude of which can induce significant plastification of the structural element. However, the large forces and strains within the element may not be reflected in the deformation of the structure as the deflection of the element is a result only of the total strain. The thermal strain of the element can further be divided into its constituents being the strains resulting from thermal expansion and thermal bowing represented by the following equation.

$$\varepsilon_{thermal} = \varepsilon_T - \varepsilon_\phi$$

A combination of these opposite thermal actions can produce differing responses of the beam element as depicted in Figure 1. A restrained expansion dominated behaviour will result in compression and a thermal bowing dominate behaviour will result in tension. There can also be a case where the two behaviours cancel each other such that the thermal strain and stress within the beam is equal to zero (Usmani et al., 2001).

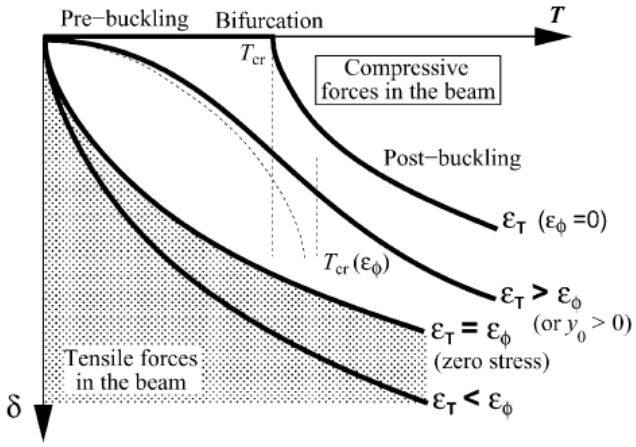


Figure 1: Plot of temperature deflection response from (Usmani et al., 2001)

During the later stages of a fire as the beam temperature rises and the strength and stiffness of the beam deteriorates, tensile catenary action can also develop in the beam element transitioning from compression and bending forces to axial tension. This was supported by experimental research conducted by (Li & Guo, 2008) which determined that the development of catenary action was generally found to occur when the beam deflection has surpassed a deflection limit of 1/20 of the span of the beam. This is also significantly influenced by the load level, with the development of catenary action accelerated by a higher loading (Kodur & Dwaikat, 2009).

The transition to catenary action also affects the response of the beam. At the early stages, an increase in axial stiffness results in larger displacements of the element, however after the development of catenary action, the deflections slow down with increased axial stiffness resulting in smaller displacements (Martinez & Jeffers, 2021). It was also found that the length of a restrained composite beam also heavily influences its failure mode during different stages of the fire. Composite beams with shorter spans tend to fail in the compressive beam-column stage where the beam is subject to both axial force and bending moment. Whilst longer beams tend to fail in the tensile catenary stage where the loads are carried through axial tension.

Additionally, the research by (Li & Guo, 2008) also found that restrained steel beams had better fire-resistant capabilities than isolated steel beams due to the compressive and tensile forces developed at different stages, however the significant compressive axial forces in the beam from thermal expansion was observed to cause local flange buckling of the steel beam.

3. Methodology

The project will involve the development and iteration of numerical models for a composite steel beam and concrete slab section subjected to thermal exposure from below. A parametric heating profile will be generated using Fire Dynamics Simulator (FDS), a Computational Fluid Dynamics (CFD) tool for simulating fire-driven fluid flow. Fire scenarios will be modelled for various building profiles using the PyroSim interface to produce temperature-time curves at the underside of the composite beam.

These thermal profiles will serve as input for Abaqus, a Finite Element Analysis (FEA) software, which will be used to construct and simulate the thermo-mechanical response of each model. The core focus of this study is to examine the influence of material nonlinearity and thermal degradation on the behaviour of the composite section under fire conditions.

The analysis will assess:

- Sectional stress distribution
- Element capacity
- Deflection profiles
- Dominant failure mechanisms

Simulations will be conducted across a range of composite beam configurations, considering variations in universal beam sizes, slab thickness and element span lengths. The numerical model shall consider a fully composite beam with complete transfer of shear forces between steel beam and concrete slab.

Output data from the Abaqus simulations will be processed using Excel to generate plots and facilitate the identification of meaningful trends, correlations, and conclusions.

The investigative project shall be divided into the following stages:

1. *Model validation*

Validation of ABAQUS modelling software through comparison to empirical data from completed experimental research. The validation process shall involve using ABAQUS models to replicate experimental testing conducted by (Wang et al., 2017) and assessing the suitability of the application for the intended research.

2. *Exposed Temperature Profile*

Generation of temperature–time data at the exposed surface of the composite beam using parametric fire curves representative of typical building occupancy profiles (e.g., commercial, industrial).

3. *Material Nonlinearity*

Investigation of the effects of material nonlinearity on composite beams under uniform heating. This stage will involve assessment of the effects of yielding comparatively to the response of the composite beam under elastic conditions. To simulate the loss of material properties as a result of nonlinear material conditions, the model shall consider reduction factors for steel yielding and concrete cracking from Eurocode 2 and 3.

4. *Thermal Degradation*

Incorporation of thermal degradation into the model and consideration of the significance of its impact at elevated temperatures in comparison to non-linear material conditions.

5. *Comparison to Usmani et al*

Comparison of results with research by (Usmani et al., 2001) which has an absence of consideration for yielding or thermal degradation of material properties.

4. Model Validation

The application of numerical modelling does not offer the same degree of credibility as empirical experimentation or post fire event data collection. However, a level of confidence can be provided as to the accuracy of the model between numerical predictions and the real world through model validation.

Model Validation is defined by (ASME, 2006) as '*the process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model*'.

This is conducted through a comparison of quantities of interest between datasets from real world experiments and their modelled counterparts. The degree of agreement between the experiment and the model prediction is viewed as the model accuracy and forms the basis for validation of the model (Borg et al., 2014).

4.1 Experimental Data

The validation process shall rely on experimental research conducted by (Wang et al., 2017) as the real world empirical data from which the numerical simulations are compared against. The research explores the fire behaviour of steel-concrete composite beams with partial shear connection. Two full-scale composite beams with 50% and 100% shear connection ratios were tested. The composite beams were heated from beneath with the steel beams protected with fire insulation. For the purposes of the model validation, only the fully composite beam is considered.

The following parameters are obtained from the details of the test specimen.

- 5260 mm long composite beam with 1250 mm wide concrete slab.
- 6 m long beam made from grade Q235 structural steel with 235 MPa yield strength.
- The H-beam had section dimensions of 320 mm x 140 mm x 8 mm x 10 mm.
- The concrete was tested to have a Young's Modulus of 21750 MPa and an average compressive strength of 21.55 MPa.
- 11 mm thick sprayed fire insulation coating applied to beam.
    - The fire insulation has a thermal conductivity, specific heat and density of 0.05 W/mK, 1100 J/kgK and 450 kg/m³ respectively.

During the test, the composite beam is subject to furnace temperatures that closely follows the ISO 834 standard fire curve, with four equivalent vertical loads of 45 kN applied along the length of the beam as shown in Figure 2.

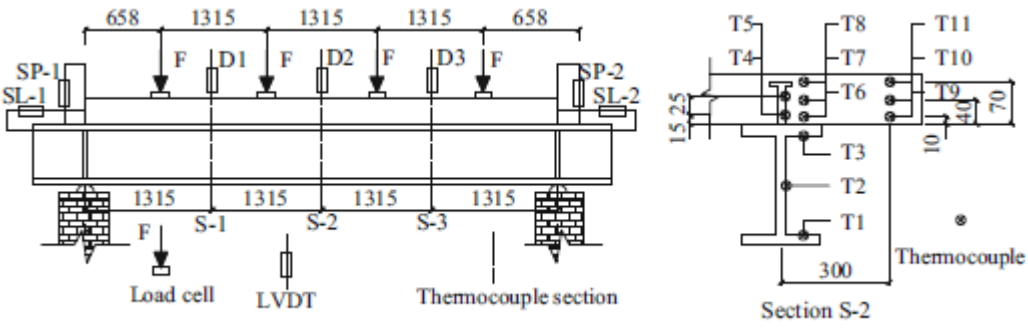


Figure 2: Location of Measurement Instruments (Wang et al., 2017)

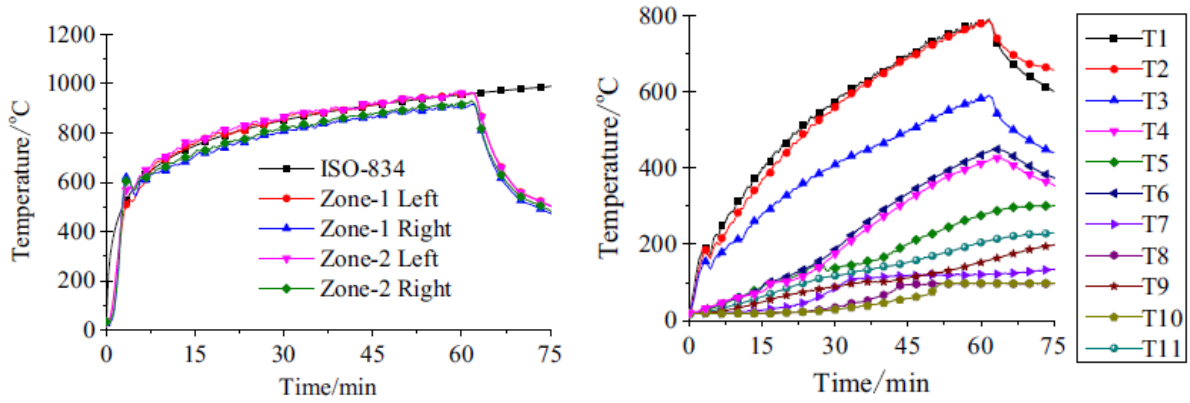


Figure 3: Temperature-Time Curve of Furnace (Left) and Thermocouples (Right) (Wang et al., 2017)

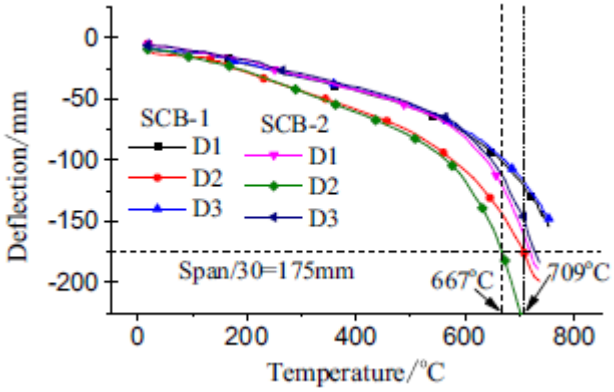


Figure 4: Vertical Displacement to Temperature at Bottom Flange (Wang et al., 2017)

Measurements of the following parameters are taken during the testing through the use of thermocouples and Linear Variable Differential Transformer (LVDTs):

- T1, T2, T3 – Steel temperatures at the bottom flange, web and top flange of the beam.
- T4, T5 – Temperature of the steel shear studs.
- T6, T7, T8 – Temperature of the slab at 10 mm, 40 mm and 70 mm above the steel beam.
- D1, D2, D3 – Vertical displacement of composite beam at different points along its length.
- SL1, SL2 – Slip displacement of slab at each end of composite beam.
- SP1, SP2 – Uplift displacement of slab at each end of composite beam.

4.2 Numerical Simulation

4.2.1. Heat Transfer

Simulation of the composite beam subject to fire was achieved in two parts. Firstly, the temperate development within the steel beam and concrete slab was calculated through a 2D sectional heat transfer model. The generated temperature profile of the composite section is shown in Figure 5.

The composite element was exposed to an ISO 834 standard fire for 60 min with decay until 75 min as per the temperature profile in Figure 3. Surfaces on the underside of the composite beam was exposed to radiative and convective heating and the top surface of the concrete was subject to ambient cooling. The 11 mm thick insulation layer is incorporated around the perimeter of the steel beam.

Temperature values at the thermocouple locations for the steel beam (T1, T2, T3) and through the depth of the concrete (T6, T7, T8) are extracted from the heat transfer model and plotted in Figure 6. Although the proportional offset between different parts of the composite beam was representative of the experimental readings, the magnitude was significantly lower than that recorded by the thermocouples with peak steel temperatures of 137°C in the model compared to 800°C in the experiment.

The discrepancy can be attributed to simplifications and assumptions of the experiment made when constructing the model and imperfections of the experimental setup which are unaccounted for in the numerical simulation. Factors responsible can include:

- Heat loss via the loading rig and structure supporting the concrete slab.
- Imperfections in the application of the insulation spray.
- Imprecise material properties of the steel and concrete used in the numerical model which are not divulged in the experimental report.
- The metallic reinforcing bar truss within the concrete slab is not entirely captured and is simplified as staggered rebars in the model. This can cause significant heat transfer through the thickness of the slab.

Notwithstanding, the temperatures of the steel beam are substantially offset from the experimental results which can suggest a major inaccuracy in the input. The likely discrepancy is understood to be the insulation layer around the steel beam which is administrated as a spray. As the insulation is applied on-situ, effective material properties may differ from its technical data. Imperfections within the insulation layer created during application can further lead to increased heat transfer.

On this basis, the model was regenerated with an increased conductivity of 5 W/mK to the insulation with the outputs shown in Figure 8. Resultant temperatures were observed to be in much closer alignment to the thermocouple measurements in Figure 3, noting that the temperature drop during the decay period is now apparent.

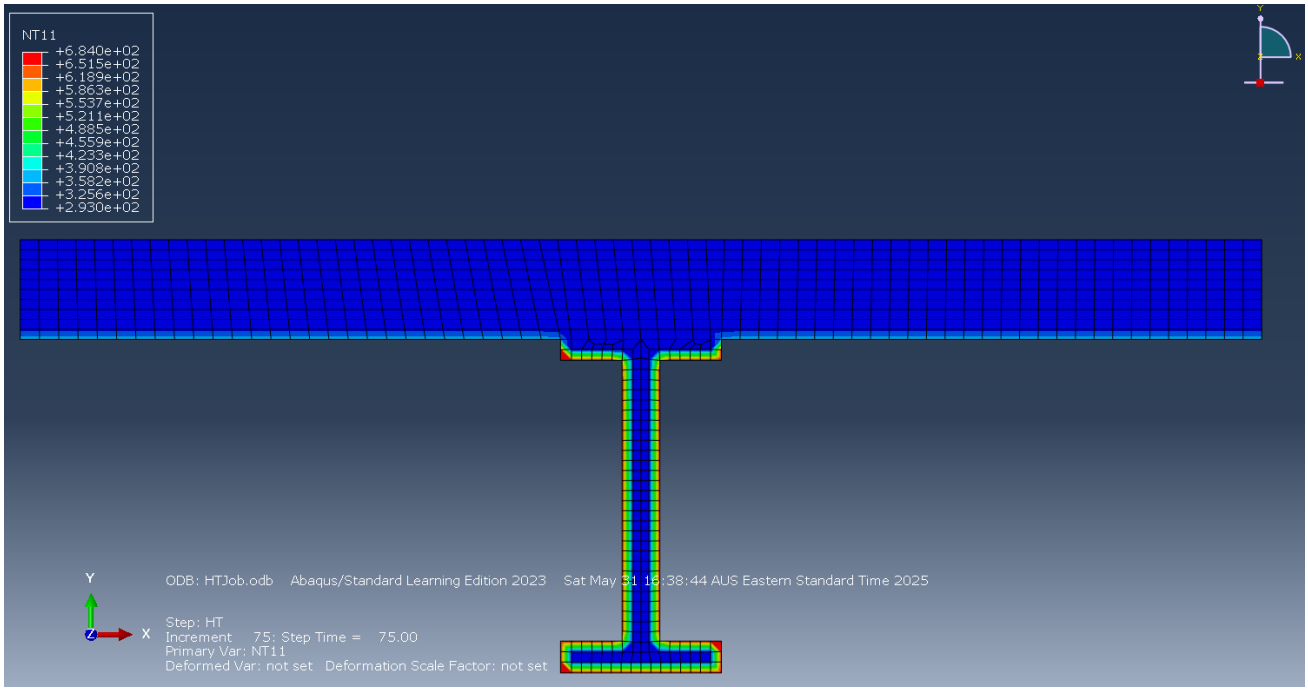


Figure 5: Temperature Profile of Heat Transfer Model – Insulation properties as specified in report

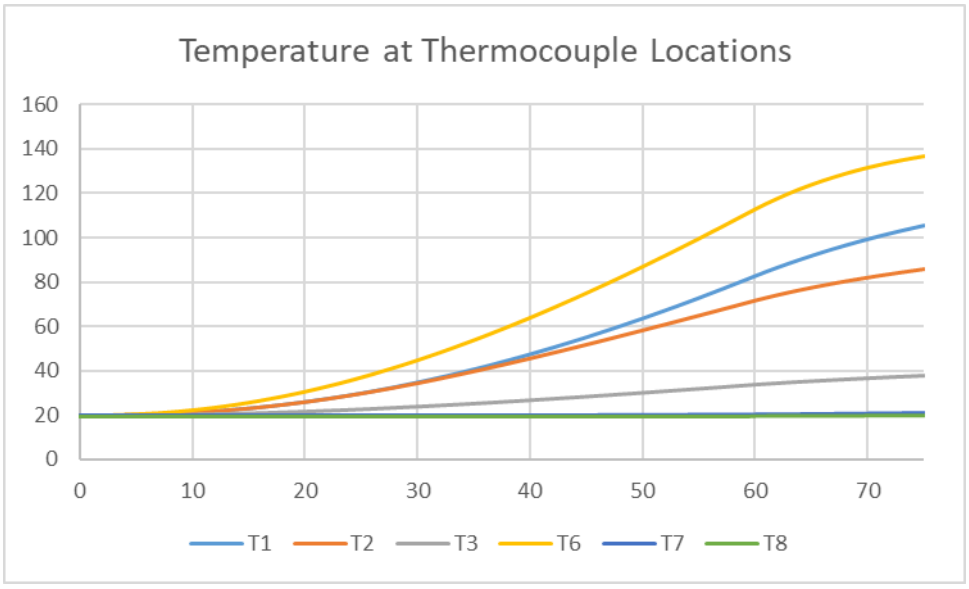


Figure 6: Nodal Temperature at Thermocouple Locations – Insulation properties as specified in report

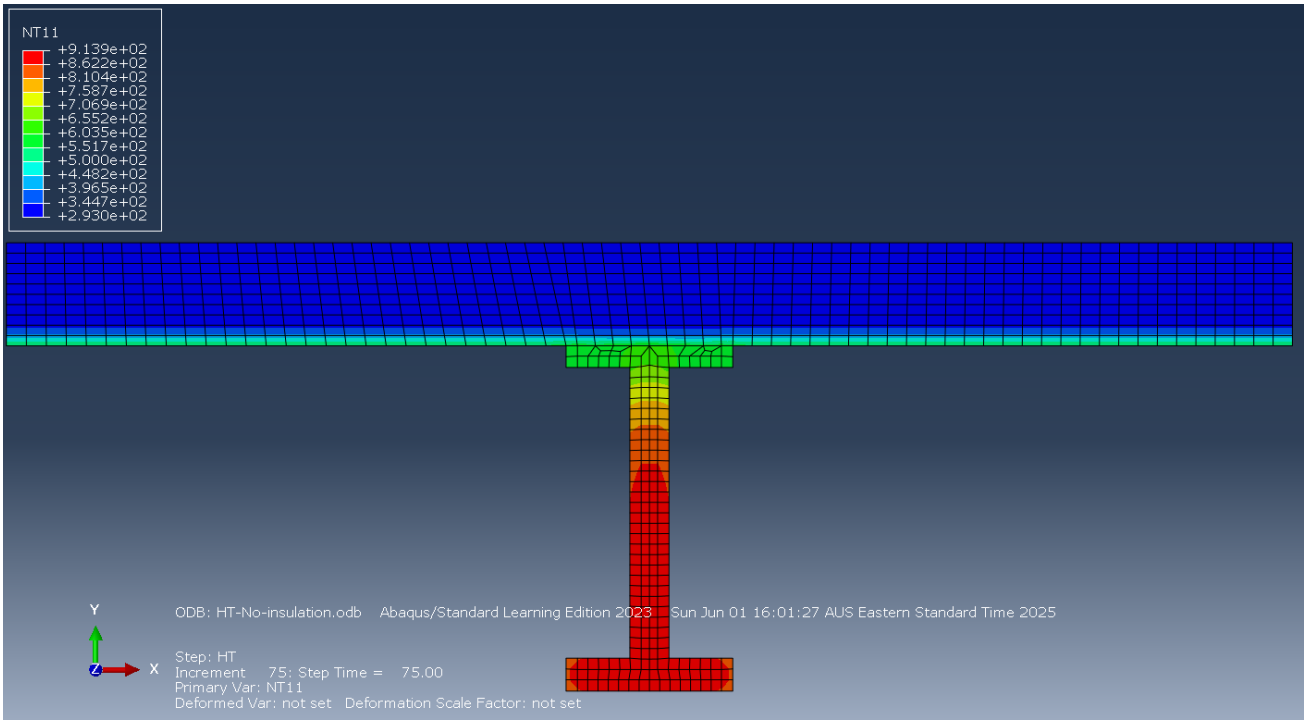


Figure 7: Temperature Profile of Heat Transfer Model – Increased insulation conductivity

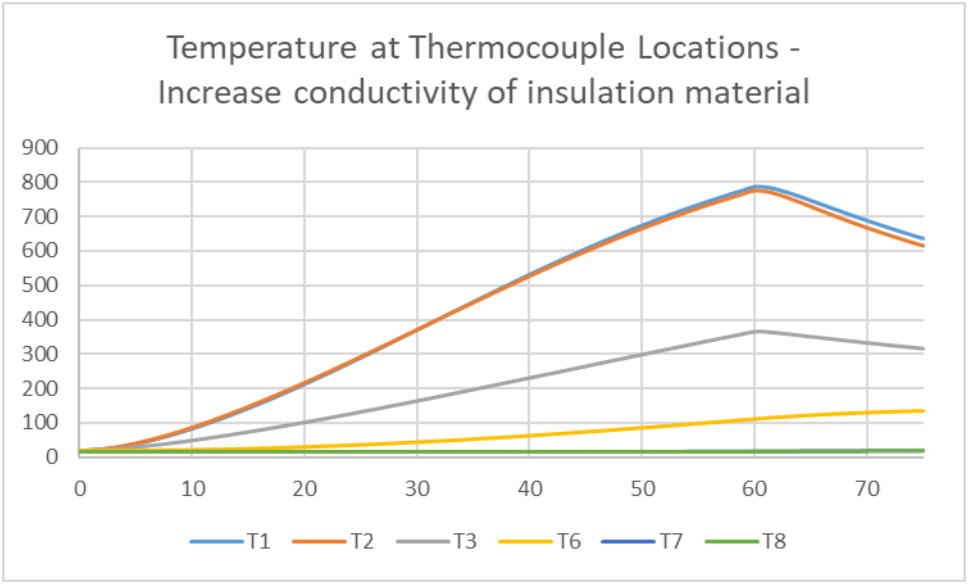


Figure 8: Nodal Temperature at Thermocouple Locations – Increased insulation conductivity

4.2.1. Thermo-Mechanical Model

A subsequent 3D thermos-mechanical model is generated using a predefined time dependent temperature profile from the heat transfer model. Due to mesh constraints permitting a maximum of 1000 node in the academic version of the software, the model was

constructed using a shell and wire framework in lieu of a solid extrusion to achieve simplified meshing. Vertical loads of 45 kN was applied to 4 points along the composite beam in the same locations as the experiment. Given the high thermal conductive of steel, a uniform temperature was applied to the steel beam and a temperature gradient was applied through the depth of concrete slab.

A contour plot of the deflection is shown in Figure 9.

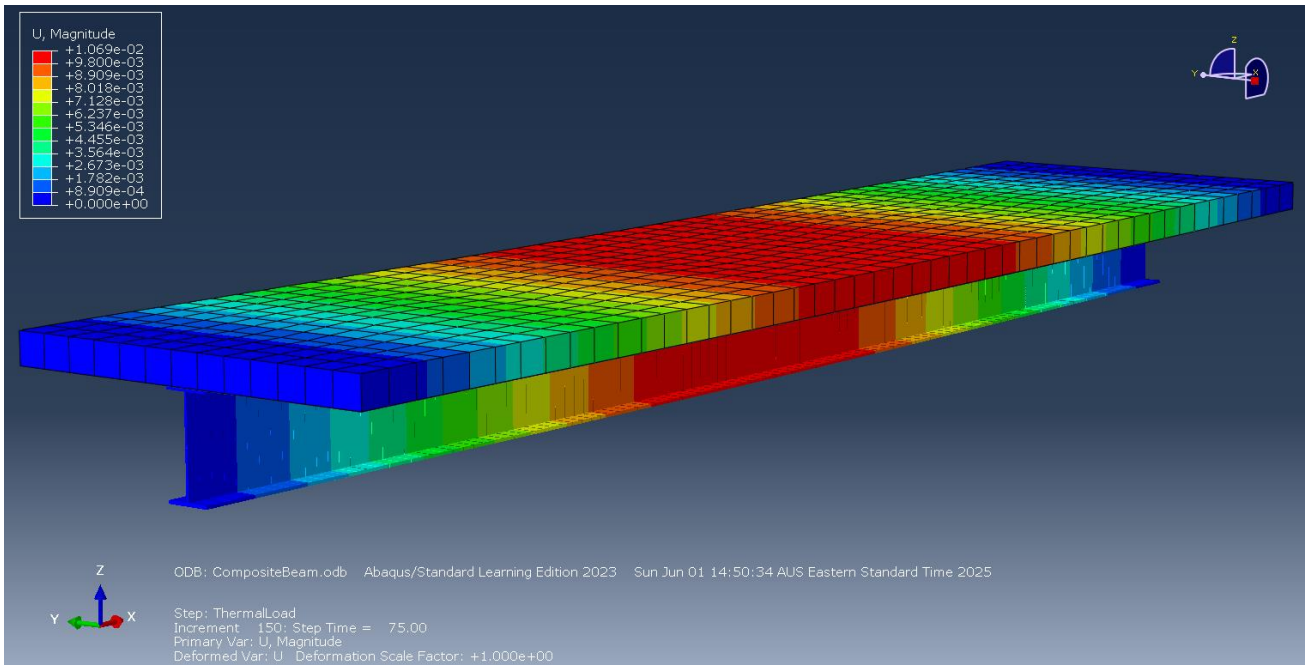


Figure 9: Deflection Contour of Thermo-Mechanical Model

Assumptions

- Reinforcing bar truss is not considered in the model
- Shear studs are considered as full shear contact between slab and beam

Challenges

Due to the limited computational limit of 1000 nodes in the academic version of ABAQUS, the model could not be generated as a 3D solid extrusion as any meaningful mesh distribution would exceed the permitted limit. Therefore, as a compromise, the model is constructed using a concrete shell element with a steel beam wire below. Shear studs are not modelled and are instead assumed to provide full shear contact between the composite elements.

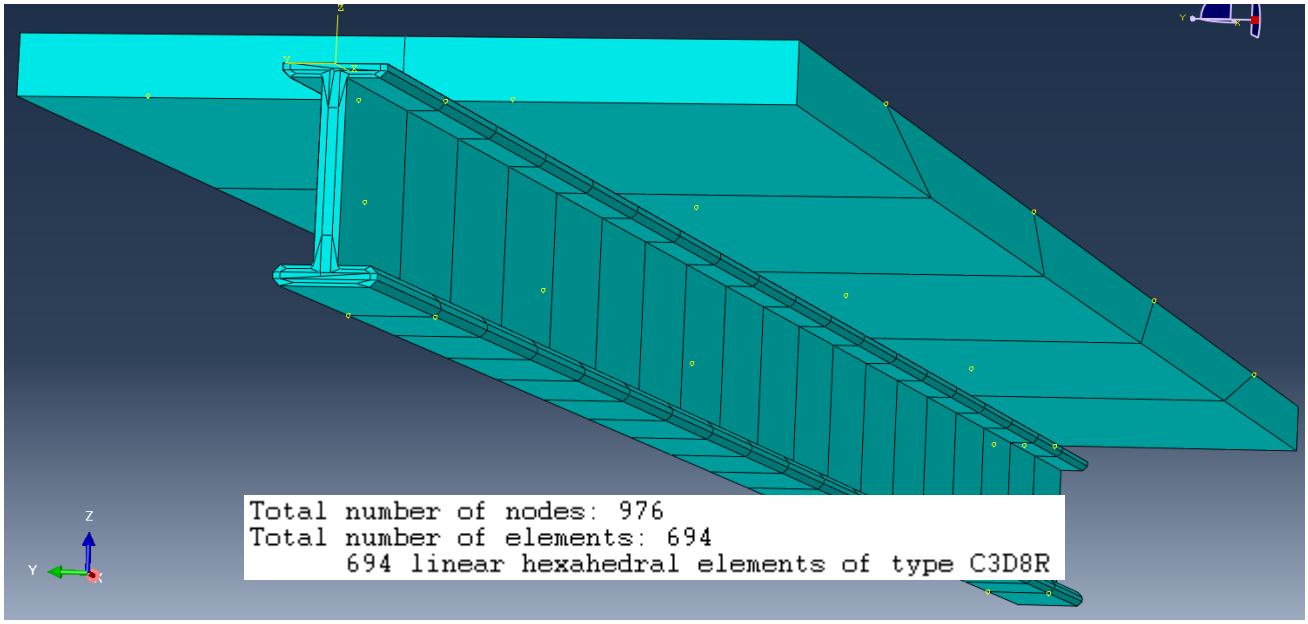


Figure 10: Mesh size of 3D model with <1000 nodes

It was found in the thermo-mechanical model, the simulation was only successful at lower temperatures (<300°C) and the model fails to run when increasing the temperature of the steel and concrete with ABAQUS unable to compute a convergent solution. This has yet to be resolved.

Job: CompositeBeam Status: Aborted

Step	Increment	Att	Severe Discon Iter	Equil Iter	Total Iter	Total Time/Freq	Step Time/LPF	Time/LPF Inc
2	1	8U	0	4	4	1	0	3.05176e-06
2	1	9U	0	4	4	1	0	7.62939e-07
2	1	10U	0	4	4	1	0	1.90735e-07
2	1	11U	0	4	4	1	0	4.76837e-08
2	1	12U	0	4	4	1	0	1.19209e-08

Log | ! Errors | ! Warnings | Output | Data File | Message File | Status File

Time increment required is less than the minimum specified

Abaqus/Standard Analysis exited with an error - Please see the message file for possible error messages if the file exists.

Figure 11: Non-convergent error when beam subject to >300°C

5. Parametric Fire

Temperature exposure of the composite beam shall be evaluated using CFD simulation of parametric fire representative of typical building occupancies. The following two scenarios are considered in contrast:

- An industrial warehouse fire with an accelerated growth rate and significant fuel loads with a lower compartment temperature due to large enclosure size.
- A commercial office fire with a slower growth rate and higher compartment temperatures due to ventilation-controlled conditions after flash over.

The following characteristics parameters are selected for each simulated fire scenario.

Table 1: FDS Input Parameters

	Commercial Office	***Industrial Facility***
Fire Growth Rate	Medium t^2 (BS9999)	Ultrafast t^2 (BS9999)
Max Heat Release Rate	350 kW/m²	2000 kW/m²
Fuel Load Density	420 MJ/m² (IFEG)	1180 MJ/m² (IFEG)
Floor Area	30 m²	5000 m²
Ceiling Height	3 m	6 m

Bibliography

ASME. (2006). *Guide for Verification and Validation in Computational Solid Mechanics*.

Bailey, C. G., White, D. S., & Moore, D. B. (2000). The tensile membrane action of unrestrained composite slabs simulated under fire conditions. *Engineering Structures*, *22*.

Borg, A., Husted, B. P., & Njå, O. (2014). The concept of validation of numerical models for consequence analysis. *Reliability Engineering and System Safety*, *125*, 36-45. https://doi.org/https://doi.org/10.1016/j.ress.2013.09.009

Engelhardt, M. D., & Morovat, M. A. (2022). Directions in structural-fire safety design for steelbuildings. *Japan Architectural Review*, *5*(1). https://doi.org/0.1002/2475-8876.12250

Gillie, M., Usmani, A. S., & Rotter, J. M. (2002). A structural analysis of the Cardington British Steel Corner Test. *Journal of Constructional Steel Research*, *58*, 427–442.

Kodur, V. K. R., & Dwaikat, M. M. S. (2009). Response of steel beam–columns exposed to fire. *Engineering Structures*, *31*.

Kowalski, R., Głowacki, M., & Wróblewska, J. (2018). Thermal bowing of reinforced concrete elements exposed to non-uniform heating. *Archives of Civil Engineering*, *LXIV*(4).

Lamont, S. (2001). *The Behaviour of Multi-storey Composite Steel Framed Structures in Response to Compartment Fires* University of Edinburgh].

Li, G., & Guo, S. (2008). Experiment on restrained steel beams subjected to heating and cooling. *Journal of Constructional Steel Research*, *64*.

Martinez, J., & Jeffers, A. E. (2021). Analysis of restrained composite beams exposed to fire. *Engineering Structures*, *234*.

Orabi, M. A., Jiang, L., Usmani, A., & Torero, J. (2020). The Collapse Of World Trade Center 7: Revisited. 11th International Conference on Structures in Fire (SiF2020),

Rotter, J. M., Sanad, A. M., Usmani, A. S., & Gillie, M. (1999). *Structural Performance Of Redundant Structures Under Local Fires* Interflam'99, Edinburgh.

SFPE. (2016). *SFPE Handbook of Fire Protection Engineering* (5 ed.). Springer.

Usmani, A. S. (2003). Understanding the Response of Composite Structures to Fire. *Engineering Journal*, *42(2)*, 83-98. https://doi.org/https://doi.org/10.62913/engj.v42i2.849

Usmani, A. S., Chung, Y. C., & Torero, J. L. (2003). How did the WTC towers collapse: a new theory. *Fire Safety Journal*, *38*. https://doi.org/doi:10.1016/S0379-7112(03)00069-9

Usmani, A. S., Rotter, J. M., Lamont, S., Sanad, A. M., & Gillie, M. (2001). Fundamental principles of structural behaviour under thermal effects. *Fire Safety Journal*, *36*(8), 721–744. https://doi.org/https://doi.org/10.1016/S0379-7112(01)00037-6

Wang, W., Huang, G., Li, G., & Engelhardt, M. D. (2017). Behavior of Steel - Concrete Partially Composite Beams Subjected to Fire - Part 1: Experimental Study. *Fire Technology*, *53*, 1039-1058.

Appendix A – Future Work Gantt Chart

